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universidade de aveiro
departamento de eletrónica,
telecomunicações e informática

Adding OpenTelemetry to nopCommerce

Software Architectures

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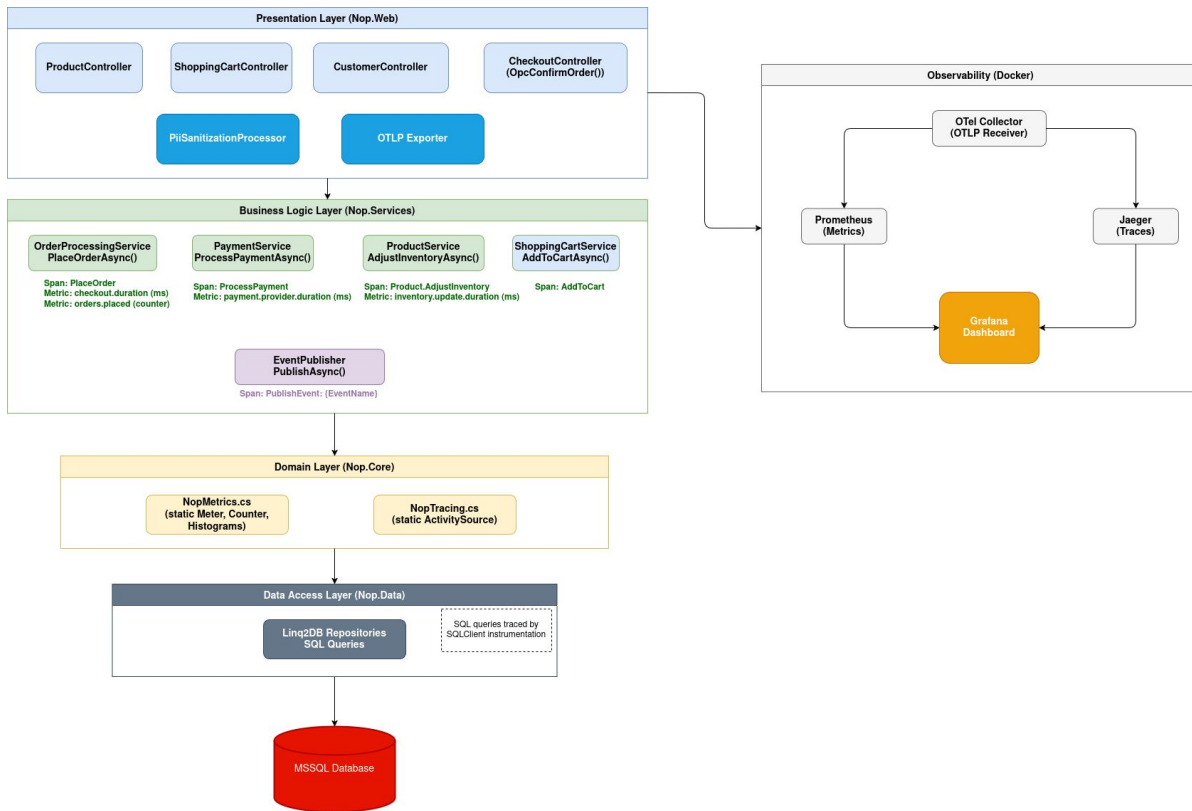
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Architecture



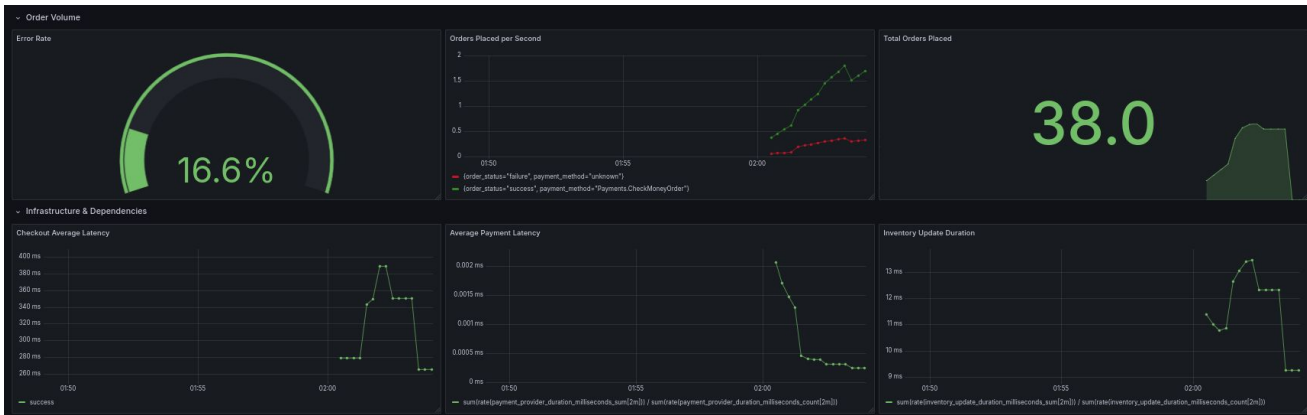
Metrics Implemented

- **orders.placed**: counter that tracks the total volume of successful and failed orders
- **checkout.duration**: measures the latency of the entire **PlaceOrderAsync** operation
- **inventory.update.duration**: tracks the performance of database inventory adjustments
- **payment.provider.duration**: measures the latency introduced specifically by the process payment operation

Privacy Strategy and Tradeoffs

- Included only non-PII business information, like Customer IDs, Payment Methods and Total Order Amount
- Explicitly excluded personal data like emails, billing addresses and credit card information
- Also redacted possible personal data that could appear in auto generated SQL traces
- Used a PII Sanitization Processor at the SDK level, before traces reached the OTLP exporter, meaning that no sensitive data left the application

Dashboard



Distributed Traces

Add Product to Cart Traces

Trace ID	Trace name	Start time	Duration
f1f6be0f1f6a26a394055bdc5f0dc6	NopCommerce.Web: POST /addproductcar	2026-03-24 02:02:38	60.0 ms
b2870f076a1597f8b8da6521f6a6352	NopCommerce.Web: POST /addproductcar	2026-03-24 02:02:38	59.8 ms
f6ad4d97837f8e9074f8d8a9c0803f5	NopCommerce.Web: POST /addproductcar	2026-03-24 02:02:38	61.3 ms
0bd1fca3ac94d9504bdad99c0101762	NopCommerce.Web: nopcommerce-db	2026-03-24 02:02:48	61.4 ms
435a78186a3f9b6b8774d06a7f7a4a7	NopCommerce.Web: POST /addproductcar	2026-03-24 02:02:37	138 ms
f4c1661a305c98ef7b0c47d8707231a	NopCommerce.Web: POST /addproductcar	2026-03-24 02:02:37	145 ms

Process Payment Traces

Trace ID	Trace name	Start time	Duration
abc3b657229b64caaa2b67cb33bab	NopCommerce.Web: POST /checkout/OpccCo	2026-03-24 02:02:27	359 ms
5e7321fab064c713f8892a5d5502c0e9	NopCommerce.Web: nopcommerce-db	2026-03-24 02:01:58	151 ms
3caaa3a0d0a9d63b63b6839360737	NopCommerce.Web: nopcommerce-db	2026-03-24 02:01:58	176 ms
51aadab4476e6b0c7b5b583bc0d02d8	NopCommerce.Web: nopcommerce-db	2026-03-24 02:01:58	291 ms
ea0b2bad32376813d0f003947993e	NopCommerce.Web: PublishEvent: TaxTotalC	2026-03-24 02:01:37	268 ms
f0a02b04a94d69d4258250b0c86d830f	NopCommerce.Web: PublishEvent: TaxRateC	2026-03-24 02:01:37	304 ms

Place Order Traces

Trace ID	Trace name	Start time	Duration
abc3b657229b64caaa2b67cb33bab	NopCommerce.Web: POST /checkout/OpccCo	2026-03-24 02:02:27	359 ms
76d42af0056585c58d958d72a7a3dcd	NopCommerce.Web: POST /checkout/OpccCo	2026-03-24 02:02:07	310 ms
bdb4497aec18af5c657476a4a9d6c7e	NopCommerce.Web: POST /checkout/OpccCo	2026-03-24 02:01:47	250 ms
28899a8a048b948f20a35020fab00c	NopCommerce.Web: POST /checkout/OpccCo	2026-03-24 02:01:27	512 ms
978da4f55ada9efec4d260ba37f03ec	NopCommerce.Web: POST /checkout/OpccCo	2026-03-24 02:01:17	651 ms
529bf3d0f982693047fba074af222	NopCommerce.Web: POST /checkout/OpccCo	2026-03-24 02:01:17	514 ms

Confirm Order Traces

Trace ID	Trace name	Start time	Duration
406d250828c7b0f7a80d95a34d8b6f51	NopCommerce.Web: POST /checkout/OpccCo	2026-03-24 02:02:28	15.8 ms
abc3b657229b64caaa2b67cb33bab	NopCommerce.Web: POST /checkout/OpccCo	2026-03-24 02:02:27	359 ms
76d42af0056585c58d958d72a7a3dcd	NopCommerce.Web: POST /checkout/OpccCo	2026-03-24 02:02:07	310 ms
2a45c06da4677fcb9932a8d4430547	NopCommerce.Web: POST /checkout/OpccCo	2026-03-24 02:01:48	131 ms
bdb4497aec18af5c657476a4a9d6c7e	NopCommerce.Web: POST /checkout/OpccCo	2026-03-24 02:01:47	250 ms
28899a8a048b948f20a35020fab00c	NopCommerce.Web: POST /checkout/OpccCo	2026-03-24 02:01:27	512 ms



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Questions?

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